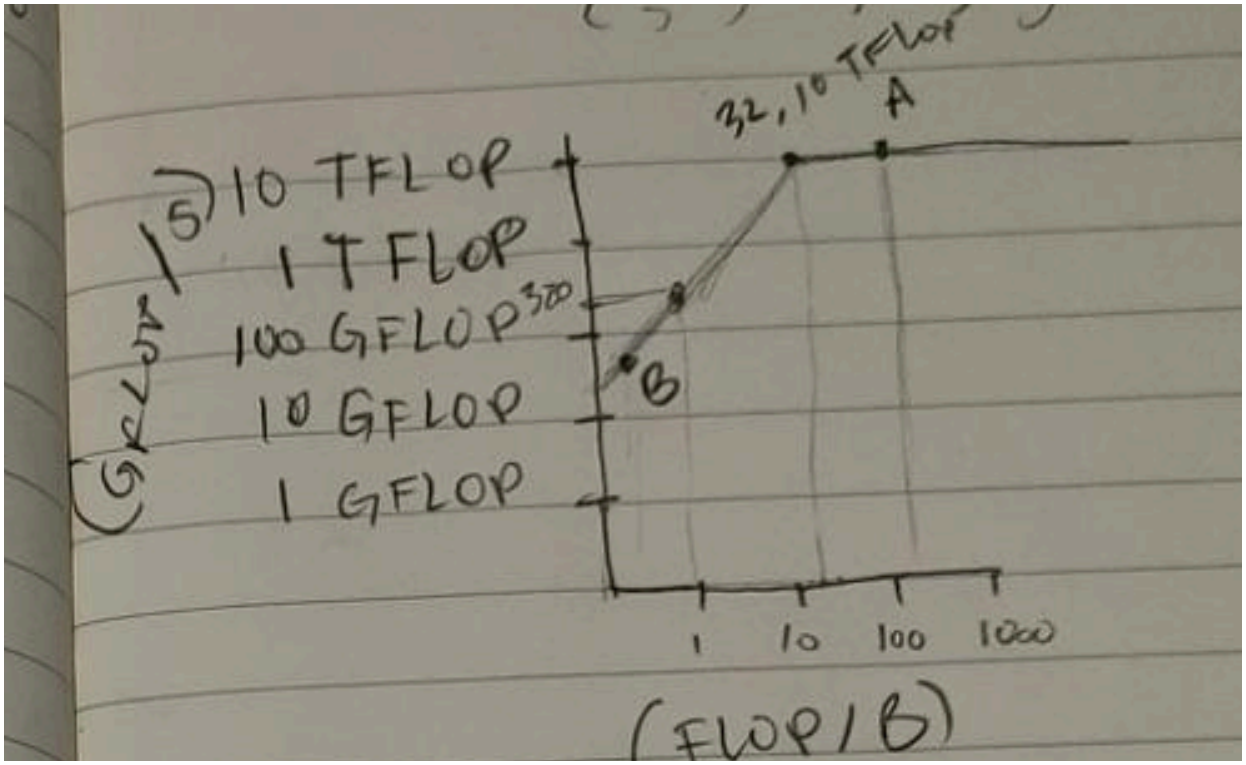
1. Roofline diagram



2. Kernel A GEMM calculation

2N^3 FLOPs = 2.147 GFLOP
12N^2 bytes = 12.583 MB

170.6 FLOPs/byte

3. Kernel B Vector addition

N FLOPs = 4194304
3N*4 bytes = 12 * 4194304

1/12 = 0.083 FLOPs/byte

4.
   a. Kernel A is compute bound, while kernel B is memory bound
   b. Attainable performance for kernel A is 10,000 GFLOP/s because it is compute bound.
      Attainable performance for kernel B is 0.083 FLOP/byte * 320 GB/s = 26.66 GLOP/s
   c. Increasing the amount of compute units (ALUs) would help kernel A.
      Increasing memory bandwidth so that the algorithm is not just sitting idle waiting for more data would help kernel B.